

User Acceptance Testing (UAT) Documentation

IITIIM Job Assistant Platform

Milestone 1: User Registration & Google OAuth

1. Description & Goal

Provide secure onboarding and authentication for site visitors. Users can register with credentials or sign up via Google OAuth. To maintain exclusivity, all new accounts are registered in a pending state and require manual review by an administrator before they can access platform features.

2. Codebase Mapping

OAuth Callback Route

- File: app.py
- Route: `@app.route("/auth/google/callback")`
- Implements Google's OAuth2 flow. It exchanges the authorization code for an access token, calls Google's userinfo API (<https://www.googleapis.com/oauth2/v2/userinfo>) to retrieve the user's profile details (name, email, avatar), and redirects back to the frontend with query parameters (`oauth_ok=1&name=...&email=...`) to pre-fill the registration form.

Standard Registration Route

- File: app.py
- Route: `@app.route("/api/signup", methods=["POST"])`
- Accepts registration details (name, last_name, email, password, linkedin_url, mobile_number). Validates that the LinkedIn URL is a valid LinkedIn profile and passes the request to the database layer.

Database Operations & Hashing

- File: database.py
- `save_user()`: Inserts standard and OAuth registrations into the users table. The password is dynamically hashed using SHA-256 with a salt before storage. Initializes `status = 'pending'` and sets the `submitted_at` timestamp.
- `validate_linkedin_url()`: Validates user-submitted LinkedIn URLs using regular expressions to ensure they point to legitimate profiles.

Frontend Integration

- Files: `IITIIMJobAssistant_v3.html`, `config.js`
- Google client-side OAuth button triggers authorization redirect. URL-parameter parsing logic extracts Google profile data to populate the signup fields automatically.

3. Database Schema (users.db)

```
CREATE TABLE IF NOT EXISTS users (  
  id          INTEGER PRIMARY KEY AUTOINCREMENT,  
  name        TEXT      NOT NULL,  
  last_name   TEXT      NOT NULL,  
  email       TEXT      NOT NULL UNIQUE,  
  password_hash TEXT    NOT NULL, -- SHA-256 hashed password  
  linkedin_url TEXT      DEFAULT '',  
  mobile_number TEXT     DEFAULT '',  
  avatar      TEXT      DEFAULT '',  
  status      TEXT      DEFAULT 'pending', -- pending | approved | rejected  
  reject_reason TEXT     DEFAULT '',  
  submitted_at TEXT      NOT NULL  
);
```

4. Verification & Testing Procedure

Google OAuth Sign Up

1. Navigate to the registration portal. Click Sign up with Google.
1. Complete the Google authorization window.
1. Verify that you are redirected back to the sign up form with your name and email pre-filled.

Standard Registration Validation

1. Attempt to register with an invalid LinkedIn URL (e.g., google.com). Verify that the API rejects it with a 400 Bad Request.
1. Submit a valid registration. Verify a 201 Created response.
1. Inspect users.db and verify the password is not stored in plaintext (is a 64-character SHA-256 hex digest) and status is 'pending'.